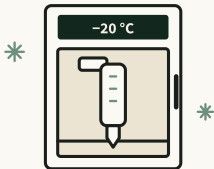




Exhaled Breath Condensate (EBC) Collection

Five steps from the pre-cooled condenser to the analysis laboratory. For nurses and physicians.

1



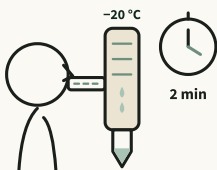
Retrieve the pre-cooled condenser

Take the $-20\text{ }^{\circ}\text{C}$ breath condenser from the freezer in the sample collection room. Check that the Eppendorf tube is seated at the bottom of the condensation tube, fit a clean mouthpiece and start at once while the tube is cold.

✓ Gloves on. Label the tube: participant ID · date · time.

$-20\text{ }^{\circ}\text{C}$
 PRE-COOLED
 CONDENSER

2



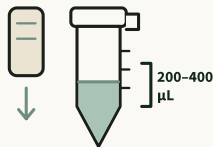
Exhale strongly for 2 minutes

The participant breathes in through the nose and exhales strongly and steadily through the mouthpiece into the $-20\text{ }^{\circ}\text{C}$ tube for 2 timed minutes. Keep the device upright so the condensate runs down into the Eppendorf tube.

✓ No coughing, talking or saliva into the mouthpiece.

2 min
 STRONG, STEADY
 EXHALATION

3



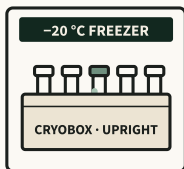
Recover the condensed sample

Detach the Eppendorf tube from the bottom of the condenser. Expected yield is about 200–400 μL of condensate, which may be partly frozen. Cap the tube immediately and record the approximate volume on the sample form.

✓ Low volume? Record it. Never top up or pool samples.

200–400
 μL
 EXPECTED YIELD

4



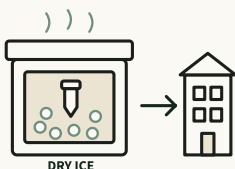
Store the Eppendorf tube at $-20\text{ }^{\circ}\text{C}$

Place the capped, labelled tube upright in the cryobox in the $-20\text{ }^{\circ}\text{C}$ freezer immediately after collection and log its position and time. Return the condenser to the freezer to re-cool for the next participant.

✓ Once frozen, it stays frozen: no thaw and refreeze.

$-20\text{ }^{\circ}\text{C}$
 STORE AT ONCE,
 UPRIGHT

5



Ship to Gipfel Life Sciences on dry ice

Send frozen samples to the Gipfel Life Sciences laboratory in an insulated box packed with dry ice so they stay at $-20\text{ }^{\circ}\text{C}$ or colder throughout. Enclose the sample manifest (ID, date, volume) and notify the lab before dispatch.

✓ Samples must never thaw in transit.

Dry ice
 $-20\text{ }^{\circ}\text{C}$ OR
 COLDER IN
 TRANSIT

CRITICAL POINTS

- * **Cold chain.** Condenser pre-cooled, tube frozen at once, dry ice in transit.
- 🗑 **Clean sample.** New mouthpiece each time; no saliva, no coughing.
- 🔗 **Traceability.** Label and log every tube: ID, date, time, volume.

SAMPLE FORM

PARTICIPANT ID	DATE · START TIME
VOLUME, μL	POSITION · INITIALS

Before your breath sample

Food, drinks and smoke change the molecules in your breath for a while, so please follow these simple rules. Sampling is painless and takes about 2 minutes.

COUNTDOWN TO YOUR SAMPLE

2 hours before

✗ Last food. Nothing to eat from now on, not even gum or sweets. No smoking and no sexual intercourse from now on.

1 hour before

✗ Last drink other than water. No coffee, tea, juice, milk, soft drinks or alcohol. Still water is fine at any time.

Sample time

💧 Rinse your mouth with water, sit down, then breathe out steadily through the mouthpiece for 2 minutes.

PLEASE AVOID

- ✗ **Eating** for 2 hours before sampling, including chewing gum, sweets and lozenges.
- ✗ **Any drink except water** for at least 1 hour before sampling: no coffee, tea, juice, milk, soft drinks or alcohol.
- ✗ **Smoking, vaping or nicotine products** for at least 2 hours before sampling, ideally not at all that day.
- ✗ **Mouthwash, throat sprays or lozenges** within 1 hour before sampling.
- ✗ **Sexual intercourse** for at least 2 hours before sampling.
- ✗ **Strenuous exercise** within 1 hour before sampling: a gentle walk is fine.

PLEASE DO

- ✓ **Drink still water** as usual. It does not affect the sample.
- ✓ **Rinse your mouth with water** just before the sample is taken.
- ✓ **Arrive a few minutes early** and rest seated before sampling, so your breathing is calm.
- ✓ **Breathe in through your nose** and out steadily through the mouthpiece for the full 2 minutes.
- ✓ **Take your usual medication** unless your physician has told you otherwise, and tell us what you took.

PLEASE TELL THE TEAM BEFORE SAMPLING IF

- you have had a cold, cough, sore throat or other respiratory infection in the last 2 weeks
- you have asthma, COPD or use an inhaler
- you smoke or vape, or have done so today
- you could not follow one of the rules above: we will note it or find a better time

WHAT HAPPENS DURING SAMPLING

You sit comfortably and breathe out through a clean mouthpiece into a cooled tube for about 2 minutes. Your breath condenses into a few drops of liquid, which are frozen and sent to the laboratory. Nothing is injected and no blood is taken. Questions? Ask the nurse or physician taking your sample.